PROJECT 1 REPORT

E-Commerce Return Analysis and Prediction Using Machine Learning

Abstract

This project analyses 34,500 e-commerce orders to identify patterns associated with product returns and predict return likelihood. Python was used for data cleaning, exploratory analysis, feature engineering and machine learning, while MySQL was used for business-oriented analysis and Power BI for visualization. Logistic Regression and Random Forest were developed and evaluated using multiple classification metrics. Return probabilities were further converted into Low, Medium and High risk categories to support proactive return management.

Introduction

Product returns are an important challenge for e-commerce businesses because they can increase operational costs, affect profitability, and reduce customer satisfaction. Understanding the factors associated with returns can help businesses identify problem areas and take preventive action. This project analyses an e-commerce dataset containing 34,500 order records to understand customer return behaviour. The analysis examines variables such as product category, region, payment method, customer age, discount, delivery time, order value, and profit margin. Machine learning techniques are then used to predict the likelihood of product returns. The resulting predictions are converted into risk categories that can help businesses prioritize potentially high-risk orders. Power BI is used to present the analytical and predictive results through an interactive dashboard.

Objectives

- Analyse the overall pattern of e-commerce product returns.
- Identify categories and customer segments associated with higher return rates.
- Examine the relationship between delivery time and returns.
- Analyse return behaviour using SQL.
- Develop machine learning models for return prediction.
- Compare Logistic Regression and Random Forest performance.
- Generate return probabilities and risk categories.
- Build an interactive Power BI dashboard.
- Provide actionable business recommendations based on the findings.

Tools Used

Python, Pandas & NumPy, Matplotlib & Seaborn, Scikit-learn, MySQL, Power BI

Methodology

Data Acquisition → Cleaning → EDA → SQL Analysis → Feature Engineering → ML Preparation → Model Development → Evaluation → Prediction → Risk Categorization → Power BI Dashboard.

The dataset contained 34,500 records and 20 initial variables. Feature engineering added age groups, delivery groups, order-date features and a binary return target. The data was divided into training and testing sets, and Logistic Regression and Random Forest models were trained.

Key Findings

The overall return rate was 5.52%, with 1,903 returned orders.

- Fashion had the highest category return rate (8.28%), followed by Electronics (7.30%).
- Grocery had the lowest return rate (1.31%).
- The East region had the highest regional return rate (5.91%).
- Orders taking 8+ days to deliver had the highest delivery-group return rate (6.06%).
- Customers aged 35–44 had the highest age-group return rate (5.84%).
- Returned orders had a higher average order value (204.29) than non-returned orders (168.01).
- The highest category-region return rate was Fashion in North (9.11%).

Model Interpretation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	53.78%	7.27%	62.73%	13.04%	61.66%
Random Forest	79.77%	9.17%	29.92%	14.04%	61.32%

Random Forest achieved higher accuracy, while Logistic Regression achieved substantially higher recall for returned orders. Since returns are a minority class, accuracy alone is not sufficient for model selection.

Risk Categorization

Risk Category	Orders	Actual Return Rate
Low	4,095	1.29%
Medium	25,661	5.59%
High	4,744	8.75%

The increasing return rate from Low to High risk indicates that the risk categories provide useful differentiation between orders with different observed return likelihoods.

Conclusion

The project demonstrates how SQL, Python, Machine Learning and Power BI can be combined to analyze and predict e-commerce returns. The analysis identified key return patterns, while machine learning provided probability-based risk assessment. The clear difference in actual return rates across risk categories demonstrates the potential of predictive analytics to support proactive return management and data-driven business decisions.